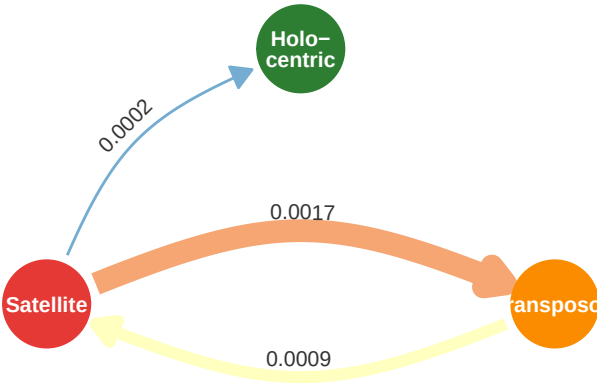


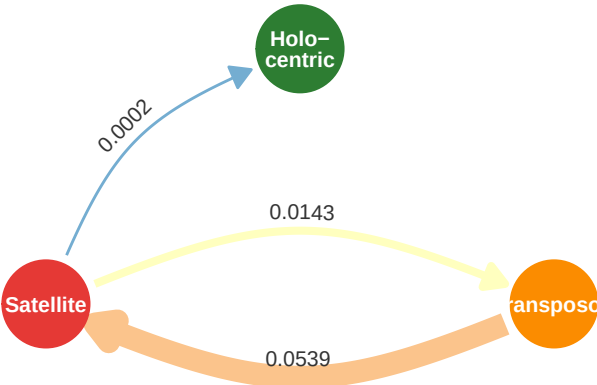
Centromere–architecture transition rates (best 3–state model per dataset, chronos–correlated)

ARD_irrevH shown for all (H irreversible; Sat→Trans vs Trans→Sat as separate rates). Bidirectional Sat↔Trans = the a–o cycle.

Full tree ... ARD_irrevH



Metazoa ... ARD_irrevH



Viridiplantae ... ARD_irrevH

